DETERMINANTS OF HBA1C: ASSESSING DEMOGRAPHIC AND DIETARY INFLUENCES USING MULTIPLE LINEAR REGRESSION

MOTIVATION

- Prediabetes affects ~37% of U.S. adults; HbA1c is a key indicator of long-term glucose regulation.
- Prior evidence on diet and HbA1c is mixed, especially in population-based samples.
  - Lower Carbohydrates → improved HbA1c in Type 2 diabetes.
  - High-protein/high-fat diets may stabilize glucose, with weaker effects in healthy individuals.
  - Soluble fibre intake → lower HbA1c.
  - Alcohol and caffeine show inconsistent associations across studies.
- **Research question: How different demographic and dietary factors are associated with HbA1c levels among U.S.citizens?**

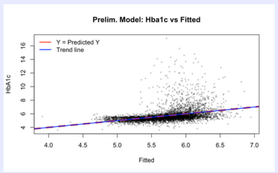
DATA COLLECTION

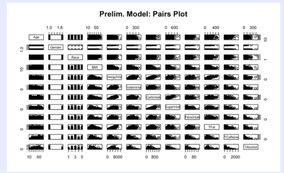
- **Source:** CDC/NCHS, National Health and Nutrition Examination Survey (NHANES), 2021–2023 for training and 2016-2017 for testing.
- **Methods:** interviews, 24-hour dietary recalls, physical exams, and laboratory standardized testing.
- **Suitability:** Representative sample; meaningful variables; independent observations.
- **Data Characteristics:**

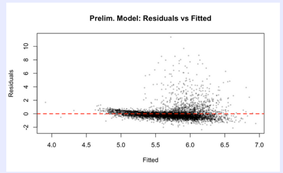
	variable	type	n	n_nonmiss	missing_pct	mean	sd	cv	skew	min	q1	median	q3	max	out_low	out_high
1	Age	numeric	5556	5556	0	49.643	20.746	0.418	−0.335	12.0	32.000	54.000	67.000	80.00	0	0
2	Gender	numeric	5556	5556	0	1.553	0.497	0.320	−0.215	1.0	1.000	2.000	2.000	2.00	0	0
3	Race	numeric	5556	5556	0	3.081	0.983	0.319	0.047	1.0	3.000	3.000	3.000	5.00	1010	1275
4	BMI	numeric	5556	5556	0	29.234	7.474	0.256	1.180	11.1	24.100	28.000	33.100	74.80	0	153
5	TEnergyIntake	numeric	5556	5556	0	1978.679	917.607	0.464	1.426	0.0	1375.000	1845.000	2435.250	10446.00	0	149
6	TProteinIntake	numeric	5556	5556	0	73.831	39.669	0.537	1.764	0.0	47.955	66.510	91.430	459.15	0	202
7	TCarbIntake	numeric	5556	5556	0	225.915	116.385	0.515	1.610	0.0	147.363	207.855	281.342	1300.94	0	165
8	TSugarIntake	numeric	5556	5556	0	96.804	68.771	0.710	2.280	0.0	51.362	82.225	125.225	953.90	0	205
9	TFibreIntake	numeric	5556	5556	0	16.057	10.390	0.647	2.034	0.0	9.000	14.100	20.500	127.30	0	197
10	TFat	numeric	5556	5556	0	83.138	46.221	0.556	1.581	0.0	51.705	75.850	105.065	505.62	0	174
11	TCaffeine	numeric	5556	5556	0	136.603	174.282	1.276	3.474	0.0	9.000	96.000	193.000	2460.00	0	220
12	TAlcohol	numeric	5556	5556	0	6.629	20.808	3.139	6.909	0.0	0.000	0.000	0.000	448.10	0	1011

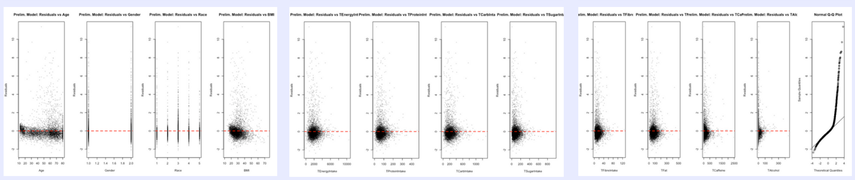
METHODS OF ANALYSIS

- **Preliminary Model:** lm(hba1c ~ Age + factor(Gender) + factor(Race) + BMI + TEnergyIntake + TProteinIntake + TCarbIntake + TSugarIntake + TFibreIntake + TFat + TCaffeine + TAlcohol + Age:Race, data = nhanes)



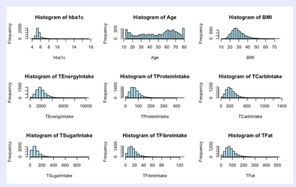




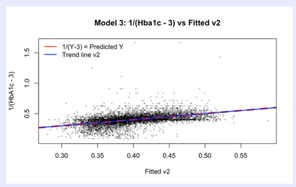


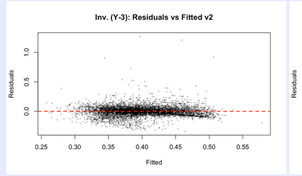
- We can see from the Fitted vs. HbA1c graph the **conditional mean response condition is satisfied** and from the pair plots that the **conditional mean predictor condition is satisfied**. We can now draw conclusions from our residual plots. Observing our residual plots, we notice the **constant variance, normality, and multicollinearity assumption are violated**. The mean of residuals is approximately 0, indicating **linearity is satisfied**.

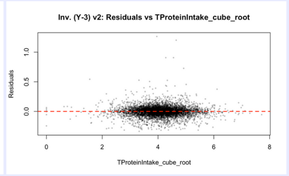
- **Middle-state model:** lm(hba1c_inv_shift ~ Age + factor(Gender) + factor(Race) + BMI_square_root_i + TEnergyIntake_cube_root + TProteinIntake_cube_root + TCarbIntake_cube_root + TSugarIntake_cube_root + TFibreIntake_cube_root + TFat_cube_root + TCaffeine_bin + TAlcohol_bin + Age:Race, data = nhanes)

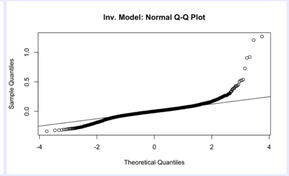


	Est Power
BMI	-0.4651
TEnergyIntake	0.4271
TProteinIntake	0.3137
TCarbIntake	0.4341
TSugarIntake	0.4138
TFibreIntake	0.2079
TFat	0.4427









- Variance Stabilization: Applied 1/(HbA1c−3) transformation on Y to reduce strong right-skew and heteroskedasticity in the response.
- Applied inverse square root transformation to BMI and cube root transformation to nutrition variables
- Recoded alcohol and caffeine to 0/1 (any vs none) because intake was very zero-inflated and extremely right-skewed.
- After these changes, constant variance looks much better; “fan” patterns in residual–fitted plots are now mild.
- Normal Q–Q plot is still S-shaped and several predictors keep moderate GVIF values, so some multicollinearity remains.

Before:

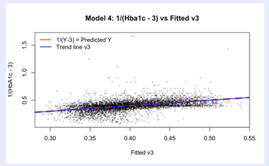
	GVIF
Age	11.477078
factor(Gender)	1.127625
factor(Race)	6.383539
BMI_square_root_i	1.093202
TEnergyIntake_cube_root	68.070843
TProteinIntake_cube_root	5.448909
TCarbIntake_cube_root	24.620524
TSugarIntake_cube_root	3.925949
TFibreIntake_cube_root	2.257695
TFat_cube_root	15.175413
TCaffeine_bin	1.166575
TAlcohol_bin	1.786652
Age:Race	16.069135

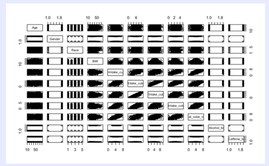
After:

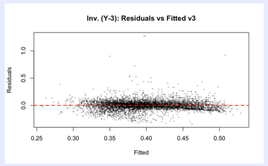
	GVIF
Age	1.236277
factor(Gender)	1.120863
factor(Race)	1.179869
BMI_square_root_i	1.092247
TProteinIntake_cube_root	2.589316
TCarbIntake_cube_root	6.560521
TSugarIntake_cube_root	3.918967
TFibreIntake_cube_root	2.184554
TFat_cube_root	2.660715
TCaffeine_bin	1.164218
TAlcohol_bin	1.058969

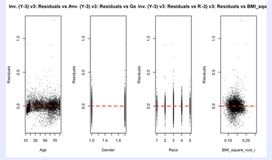
- Initially, total energy, carbs, fat, and the Age×Race interaction had very large **GVIFs**(>15), indicating severe **multicollinearity**. After removing some terms and refitting, GVIFs for the remaining predictors are low, so the final model has much less multicollinearity and more stable, interpretable coefficients.

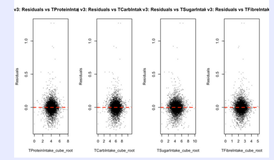
- **Final Model:** lm(hba1c_inv_shift ~ Age + factor(Gender) + factor(Race) + BMI_square_root_i + TProteinIntake_cube_root + TCarbIntake_cube_root + TSugarIntake_cube_root + TFibreIntake_cube_root + TFat_cube_root + TCaffeine_bin + TAlcohol_bin, data = nhanes)

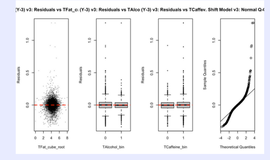






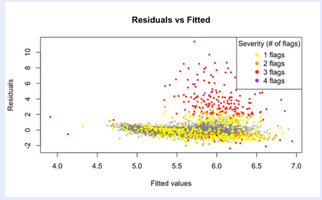




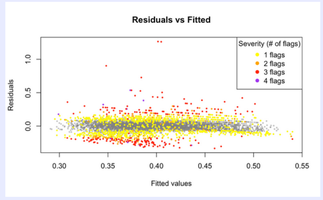


- **TEnergyIntake_cube_root** is removed as it is calculated with other nutrition variables such as TCarbIntake_cube_root, TProteinIntake_cube_root, etc.
- **Age:Race** interaction term is also removed since it has a higher VIF value than Age.

Problematic points: preliminary model



Problematic points: final model



- The final model pulls most multi-flagged **problematic points** much closer to the 0 residual line and greatly reduces the number and size of extreme outliers

- **Model Validation:** We conducted model validation using data from NHANES 2017-2018. The testing data set contains the same variables as the training data set. Variable distributions have similar trends across data sets.

Training Results:

```
Residuals:
     Min       1Q   Median       3Q      Max
-0.33486 -0.04298  0.00046  0.04207  1.26513

Coefficients:
                           Estimate Std. Error t value Pr(>|t|)
(Intercept)               3.061e-01  1.454e-02  21.044  < 2e-16 ***
Age                      -1.825e-03  6.276e-05 -29.072  < 2e-16 ***
factor(Gender)2           6.347e-03  2.494e-03   2.545  0.01094 *
factor(Race)2             1.095e-02  5.590e-03   1.959  0.05021 .
factor(Race)3             3.441e-02  4.602e-03   7.477 8.82e-14 ***
factor(Race)4             3.452e-03  5.511e-03   0.626  0.53110
factor(Race)5             1.080e-02  5.548e-03   1.946  0.05169 .
BMI_square_root_i         8.358e-01  5.435e-02  15.377  < 2e-16 ***
TProteinIntake_cube_root  2.710e-03  2.571e-03   1.054  0.29193
TCarbIntake_cube_root    -8.711e-03  2.894e-03  -3.010  0.00262 **
TSugarIntake_cube_root    6.939e-03  2.218e-03   3.129  0.00176 **
TFibreIntake_cube_root    9.215e-03  3.292e-03   2.799  0.00514 **
TFat_cube_root           -3.894e-03  2.345e-03  -1.660  0.09688 .
TCaffeine_bin1           -1.018e-03  3.498e-03  -0.291  0.77095
TAlcohol_bin1             2.773e-02  3.123e-03   8.878  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.08728 on 5541 degrees of freedom
Multiple R-squared:  0.2148,    Adjusted R-squared:  0.2128
F-statistic: 108.2 on 14 and 5541 DF,  p-value: < 2.2e-16
```

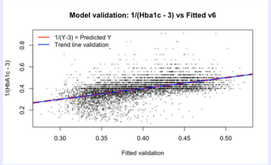
Testing Results:

```
Residuals:
     Min       1Q   Median       3Q      Max
-0.33304 -0.04180  0.00059  0.04220  0.58239

Coefficients:
                           Estimate Std. Error t value Pr(>|t|)
(Intercept)               3.337e-01  1.288e-02  25.918  < 2e-16 ***
Age                      -1.918e-03  5.479e-05 -35.010  < 2e-16 ***
factor(Gender)2           8.787e-03  2.246e-03   3.912 9.26e-05 ***
factor(Race)2             3.706e-03  4.575e-03   0.810  0.4179
factor(Race)3             2.121e-02  3.463e-03   6.125 9.67e-10 ***
factor(Race)4            -4.384e-03  3.704e-03  -1.184  0.2366
factor(Race)5            -7.447e-03  3.788e-03  -1.966  0.0494 *
BMI_square_root_i         7.151e-01  4.804e-02  14.885  < 2e-16 ***
TProteinIntake_cube_root  5.596e-03  2.252e-03   2.485  0.0130 *
TCarbIntake_cube_root    -5.659e-03  2.662e-03  -2.126  0.0336 *
TSugarIntake_cube_root    5.137e-03  2.054e-03   2.501  0.0124 *
TFibreIntake_cube_root   -3.166e-05  2.913e-03  -0.011  0.9913
TFat_cube_root           -4.008e-03  2.141e-03  -1.872  0.0612 .
TCaffeine_bin1            6.195e-04  2.941e-03   0.211  0.8332
TAlcohol_bin1             2.659e-02  2.857e-03   9.306  < 2e-16 ***
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Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

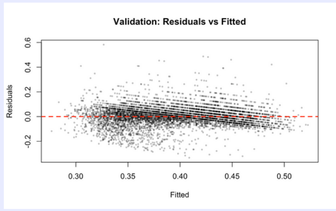
Residual standard error: 0.08052 on 5772 degrees of freedom
Multiple R-squared:   0.26,     Adjusted R-squared:  0.2582
F-statistic: 144.9 on 14 and 5772 DF,  p-value: < 2.2e-16
```

	GVIF
Age	1.185559
factor(Gender)	1.125465
factor(Race)	1.253419
BMI_square_root_i2	1.100973
TProteinIntake_cube_root	2.702131
TCarbIntake_cube_root	7.147071
TSugarIntake_cube_root	4.225595
TFibreIntake_cube_root	2.274299
TFat_cube_root	2.962361
TCaffeine_bin	1.141723
TAlcohol_bin	1.043242



- Through model validation, we confirm that our predictions are accurate beyond the testing data set and have higher probability of having true significance.

- From the result tables of the training and testing data sets, there are minimal differences (< 2 SEs) in the estimated coefficients for Age, Gender, Race2, Race4, Protein, Carb, Sugar, Fat, Caffeine, Alcohol and the intercept. Race 3, Race 5, BMI, and Fibre showed a little instability.
- All significant predictors found with the training dataset remain significant with the testing data set except for Fibre intake, indicating strong generalizability.
- The adjusted R^2 for both data sets is similar, 0.2128 (train) vs. 0.2582 (test), indicating similarly good explanation of variance.
- VIF results of testing and training models show aligned patterns, indicating similar amount of multicollinearity.
- No additional or worsening model violations.



RESULTS AND CONCLUSIONS

- The final model was statistically significant ($F(14, 5541) \approx 108$, $p < 2.2e\text{-}16$) and explained ~21% of the variance (Adjusted $R^2 = 0.213$).
- Demographic variables:
  - Age: Positive association with transformed HbA1c → older adults have higher HbA1c.
  - Gender: Women have slightly higher transformed HbA1c than men.
  - Race: Non-Hispanic White adults have significantly higher HbA1c than Mexican Americans. Other races were not significantly different.
  - BMI: Higher BMI is associated with higher HbA1c.
- Dietary variables:
  - Sugar intake, fibre intake, alcohol consumption: positively associated with transformed HbA1c.
  - Carbohydrate intake: negatively associated.
  - Protein, fat, caffeine: not statistically significant.
- All VIF values were low, suggesting multicollinearity did not bias inference.

LIMITATIONS

- **Self-reported 24-hour dietary recalls**
  - Subject to recall error and day-to-day variation.
  - Likely introduces classical measurement error → attenuates dietary effect estimates.
- **Residual non-linearity despite transformations**
  - Even after inverse-shift, cube-root, and inverse square-root transformations, several predictors still show mild curvature and residual structure.
  - Indicates that a simple linear model may not fully capture true response–predictor relationships.
- **Problematic observations**
  - Additional clinical covariates (e.g., medication) are likely missing drivers of variation.
- **Future Improvements**
  - Incorporate clinical and behavioral variables: medication use, chronic disease status, and physical activity.
  - Use richer dietary measures, such as multi-day recalls or averaged intakes.

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